

Operating Systems

Numerical notes

Part = 1

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4.4 Page Replacement Algorithms:

- 1 FIFO Page Replacement Algorithm
- 2 Belady Anomaly Page Replacement + Algorithm
- 3 Optimal Page Replacement Algorithm
- 4 LRU Page Replacement Algorithm
- 5 Second Chance Page Replacement Algorithm
- 6 Clock Page Replacement Algorithm
- 7 WS-Clock Page Replacement Algorithm

1 FIFO Page Replacement Algorithm:

Eg

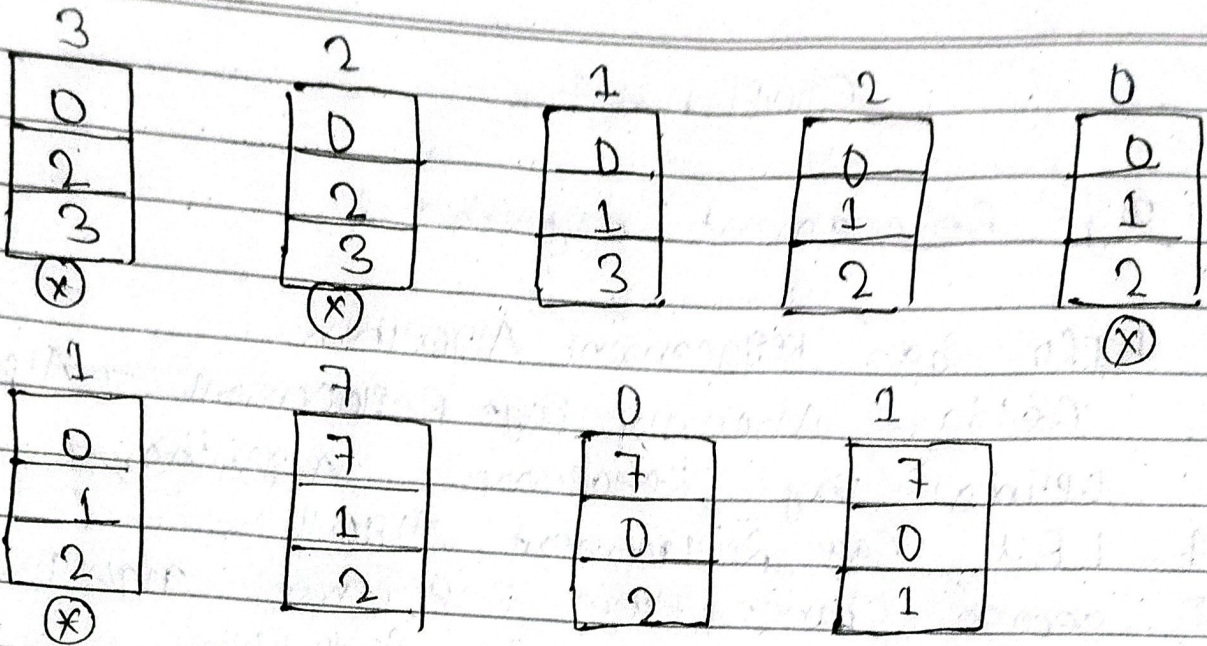
Given reference string 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1. Page frame = 3.
Find total Page faults using FIFO Page Replacement Algorithm.

Soln,

7	0	1	2	0	3
7	7	7	2	2	2
X	0	0	0	0	3
X	X	1	1	1	1

(X)

0	4	2	3	0
2	4	4	4	0
3	3	2	2	2
0	0	0	3	3

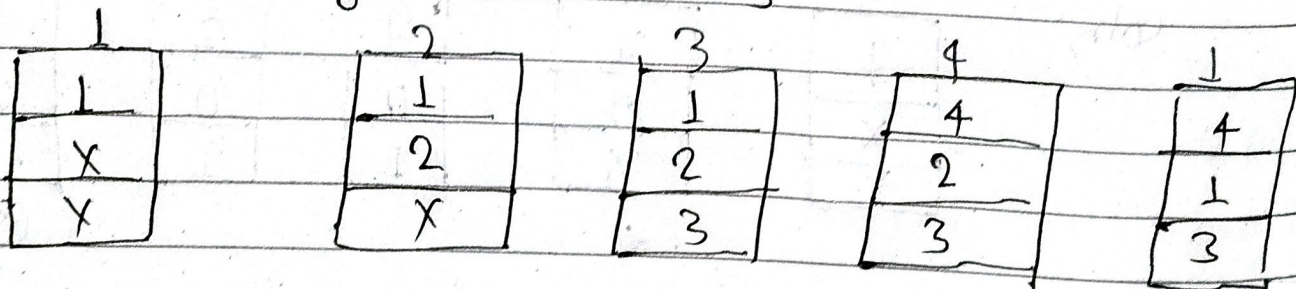


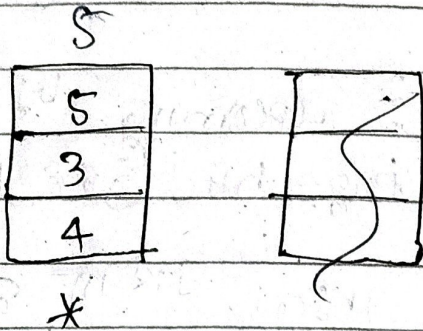
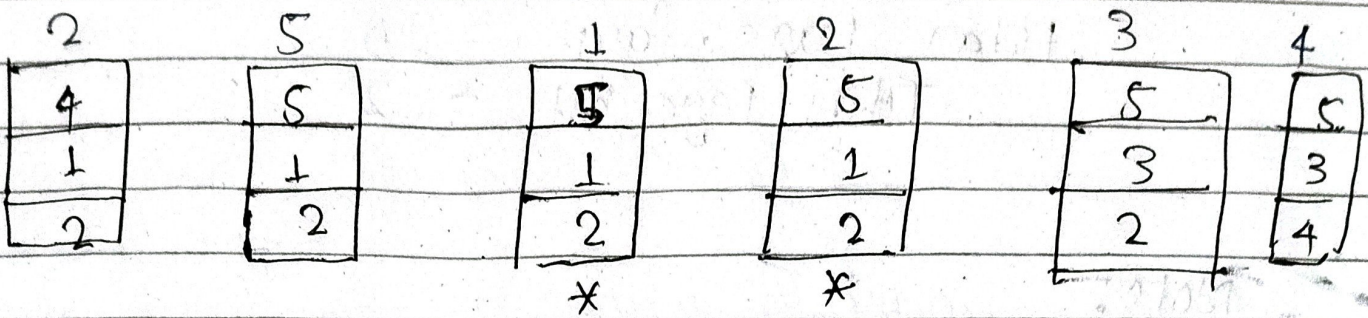
$\therefore$ total page fault = 15
 total page hit = 5

2 Belad's Anomaly Page Replacement Algorithm:

e.g Reference String: 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 with 3 using Page frame 3 & 4.

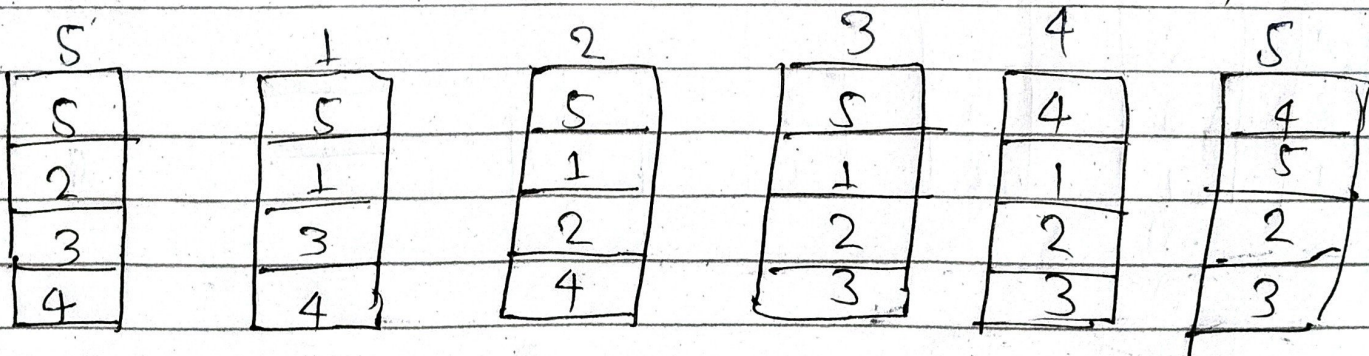
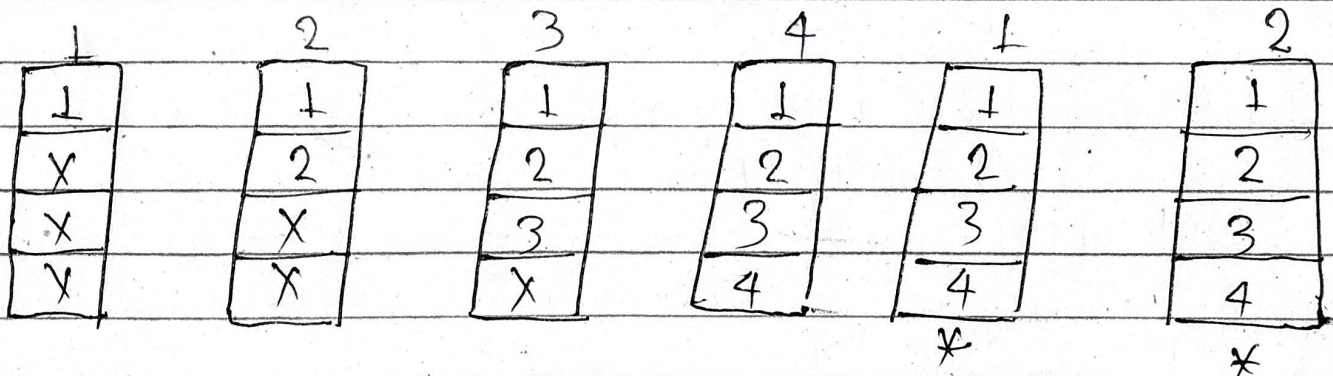
soln, page frame = 3





Total page fault = 9
 total page hit = 3

page frame = 4



total Page fault = 10

Total Page hit = 2

Note:

- 1) Box vitra Vaku number 2 upcoming હોદ્દા દ
cતો same મરતો મરતો page hit હોદ્દા, 1
- 2) હોદ્દા box vitra first મરતો insert મરતો દ
cતો 2-મરતો replace મરતો

Chapter = 4

Part = 2

3 OPTimal Page Replacement Algorithm:

Given Reference String 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1, Page frame = 3
Find total Page faults using OPTimal Page Replacement Algorithm.

Soln,

7	0	1	2	0	3
7	7	7	2	2	2
X	0	0	0	0	0
X	X	1	1	1	3
				(*)	

0	4	2	3	0	3
2	2	2	2	2	2
0	4	4	4	0	0
3	3	3	3	3	3
(*)	(*)	(*)	(*)	(*)	(*)

2	1	2	0	1	7
2	2	2	2	2	7
0	0	0	0	0	0
3	1	1	1	1	1
(*)	(*)	(*)	(*)	(*)	

0	1
7	7
0	0
1	1
(*)	(*)

total Page faults = 9

total Page hits = 11.

Note:

- i) Box vitra ke number upcoming number ke Jun table se chuthe aage replace jate.
- ii) In case upcoming no "E" se automaticly replace jate.
- iii) Replace jate box vitra same no se Page hit hote.

4 LRU (Least Recently used) Page Replacement Algorithm:

eg Given reference String 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1. Page frame = 3. Find total Page faults using LRU Page Replacement Algorithm.

7	0	1	2	0	3
7	7	7	2	2	2
X	0	0	0	0	0
X	X	1	1	1	3

(*)

0	4	2	3	0	3
2	4	4	4	0	0
0	0	0	3	3	3
3	3	2	2	2	2

(*)

(*)

2	1	2	0	1	7	0
0	1	1	1	1	1	1
3	3	3	0	0	0	0
2	2	2	2	2	7	7

(*)

(*)

(*)

(*)

1		
1		
0		
7		

(*)

Total Page Fault = 12

Total Page hits = 8

Note:

FIFO = Box uti box vitra 2120 E, H cya o
replace jata

LRU = Box vitra o 2120 compare jata
3 ota E 1 cya 2120
replace jata

S Second Chance Page Replacement Algorithm:

e.g

If we consider the reference string 2, 3, 2, 1, 5, 2, 4, 5, 3, 2, 5, 2 and number of frame allocated = 3. Find total page faults by using Second Chance Page Replacement Algorithm.

2	3	2	1	5	2
2(0)	2(0)	2(1)	2(1)	2(0)	2(1)
X	3(0)	3(0)	3(0)	5(0)	5(0)
X	X	2 X	1(0)	1(0)	1(0)
		*			⊗
2(0)	2(0)	2(1)			
X	3(0)	3(0)			
X	X	X			
		⊗			

4	5	3	2	5	2
2(0)	2(0)	3(0)	3(0)	3(0)	3(0)
5(0)	5(1)	5(0)	5(0)	5(1)	5(0)
4(0)	4(0)	4(0)	2(0)	2(0)	2(1)
	(X)			(X)	(X)

total page fault = 7

Note:

Page hits vago mat bracket mat (1) se 2
 testari at next step mat bracket mat (0)
 se 1

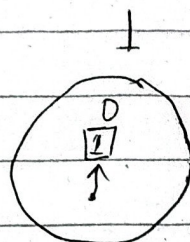
6. Clock Page Replacement Algorithm:

e.g Input reference string 1, 2, 3, 4, 3, 1, 2, 1,
 5, 4

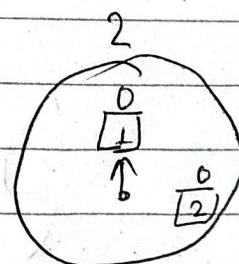
Size of memory = 3

Soln,

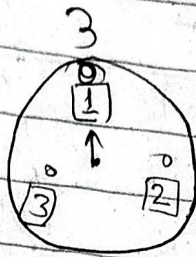
Step = 1



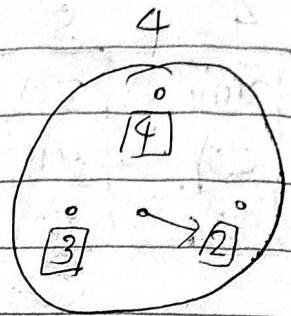
Step = 2



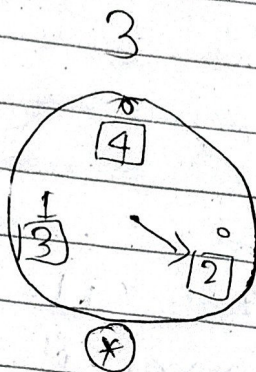
Step = 3



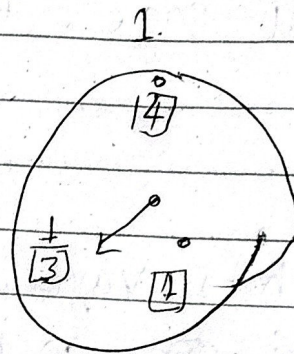
Step = 4



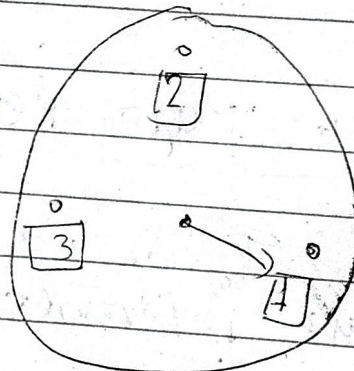
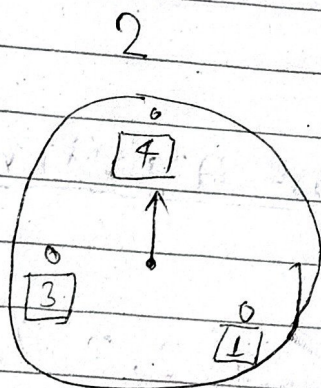
Step = 5



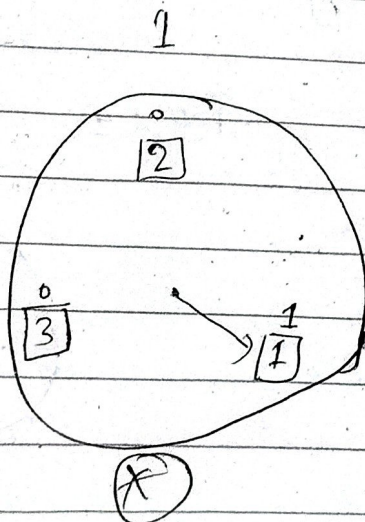
Step = 6



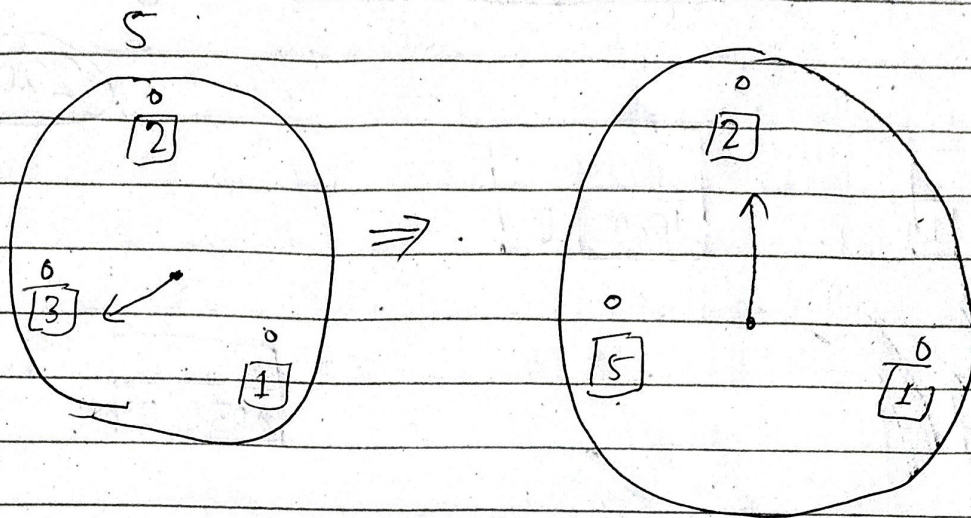
Step = 7



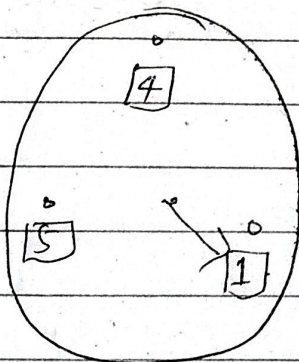
Step = 8



STEP = 9



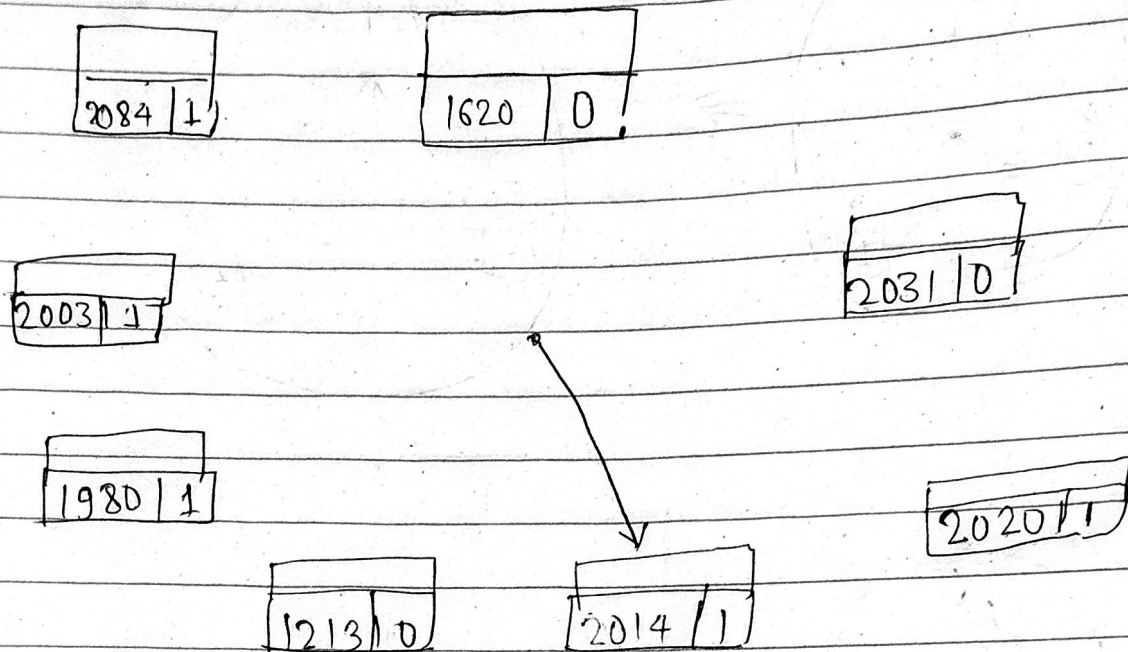
STEP = 10



Total Page Fault = 8

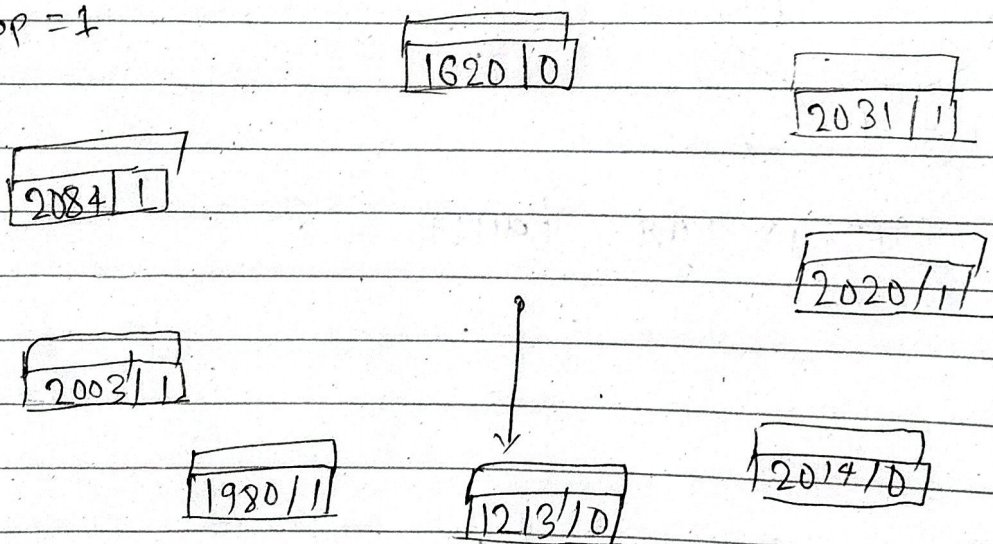
7 WS - Clock Page Replacement Algorithm:

2204



Soln,

STEP = 1



Step = 2

1620/0

2031/1

2084/1

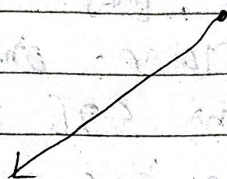
2003/1

2020/1

1980/1

2204/0

2014/0



First - Fit, Best-fit and worst-fit ?

eg Given memory partitions of 100k, 500k, 200k, 300k, 600k (in order) how would each of the first-fit, Best-fit and worst-fit algorithm place process of 212k, 417k, 112k and 426k (in order)? which algorithm make the most efficient use of memory.

Soln,

417	60k
	300k
112k	200k
212k	500k
	100k

First - fit (जहाँ ¹space मिलेगा उसे ²माँ ले लेंगे)

i) Search ~~212k~~ 212k to be fit in memory (100k fit ~~है~~ ^{नहीं} ~~so~~, 500k में 212 fit ~~मिलेगा~~ ^{है} ~~so~~, 212k, 500k को memory में ~~212~~ ^{212k} ~~ले लेंगे~~)

ii) 417, 100k में ~~है~~ ^{नहीं}, 500k में ~~है~~ ^{नहीं}, 300k में ~~है~~ ^{नहीं} ~~2~~ ² 600k में ~~212~~ ⁴¹⁷ ~~ले लेंगे~~

iii)

iii) 112K, 200K મા 2.20

iv) 426K wait કરા જો કોઈ 426K નો જગ્યા memory રાખી દેત ।

Best-fit:

(memory થોડા waste થાય)

426K	600K
212K	300K
112K	200K
417K	500K
	100K

i) space થોડા bachana paryo

ii) 212K is put in 300K થોડા space baki bachyo

iii) 417K is put in 500K થોડા space baki bachyo.

iv) 112K is put in 200K થોડા space baki bachyo.

v) 426K is put in 600K

Worst-Fit:

(memory ²⁴ 24 waste 24)

212k	600k
112k	300k
	200k
417k	500k
	100k

- i) 212 K की highest memory में 21200 K के books
- ii) 417 K " 2nd " " " के books
- iii) 112 K " 3rd " " " के books
- iv) 426 K को अगली wait जगह पर

we can see Best fit turns out to be the most efficient use of memory

(Note)

→ If frame size is not given assume it as 3.

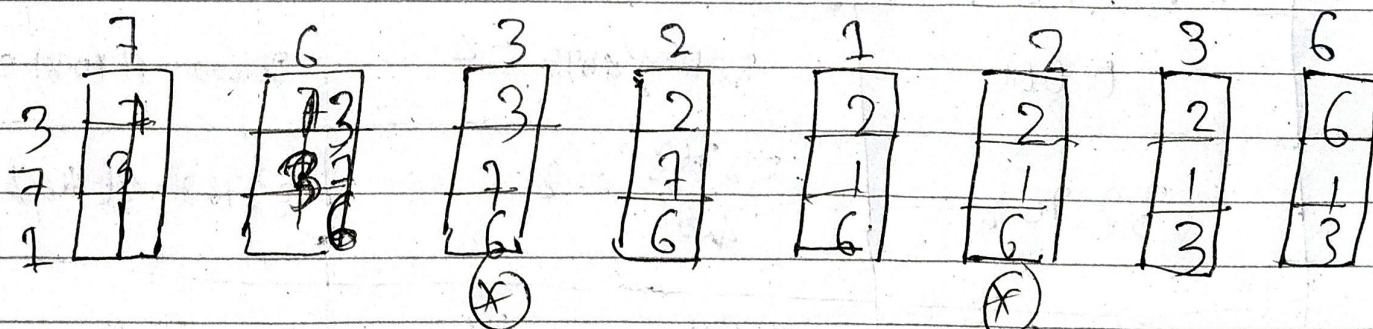
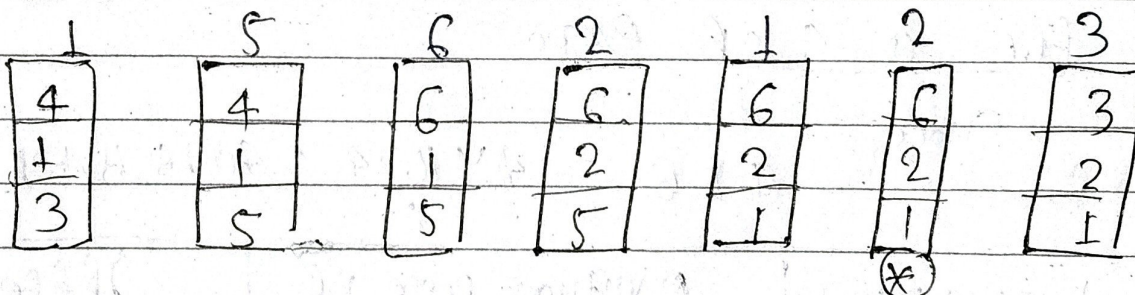
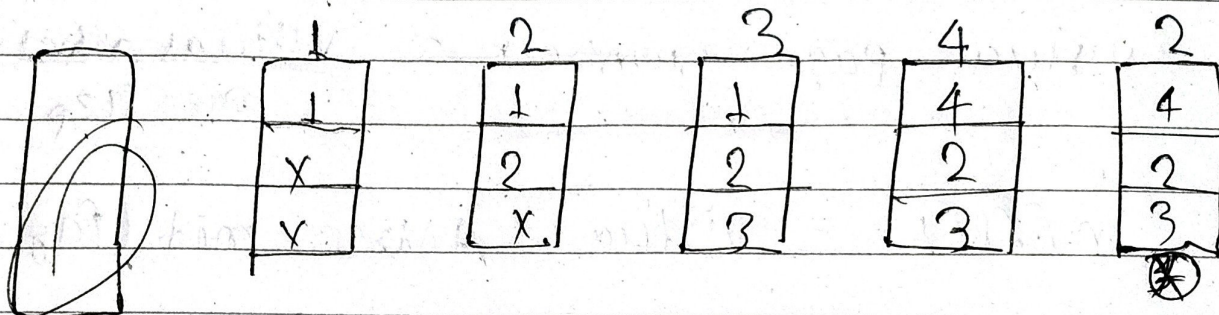
Past Year Solutions

(2082)

Given the Page reference strings 1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6.

Compute the no. of Page faults for FIFO page replacement algorithm.

Soln



Total Page Faults = 16

2082

What is Belady's anomaly? For the decimal virtual addresses 20000, 32768, 60000 compute the virtual page no and offset for 4-KB page and for an 8-KB page size.

Formula:

$$\text{Virtual page number} = \frac{\text{Virtual address}}{\text{Page size}}$$

$$\text{Offset} = \text{Virtual Address} \bmod (\text{Page size})$$

1) For a 4-KB page

Since

$$4 \text{ KB} = 4 \times 1024 = 4096 \text{ bytes.}$$

Virtual address	Virtual page no	Offset
20,000	$\frac{20000}{4096} = 4$	$20000 - 4(4096) = 3616$
32,768	$\frac{32768}{4096} = 8$	$32768 - 8(4096) = 0$
60,000	$\frac{60000}{4096} = 14$	$60000 - 14(4096) = 2556$

20,000 $\rightarrow$ virtual page number is 4 & offset is 384

32,768 $\rightarrow$ virtual page no is 8 & offset is 0.

60,000 $\rightarrow$ virtual page no is 14 & offset is 2656.

2 for a 8 KB page.

Since $8 \text{ KB} = 8 \times 1024 = 8192 \text{ bytes}$

virtual address	virtual page number	offset
20,000	$\frac{20,000}{8192} = 2$	$20,000 - 2(8192) = 3616$
32,768	$\frac{32,768}{8192} = 4$	$32,768 - 4(8192) = 0$
60,000	$\frac{60,000}{8192} = 7$	$60,000 - 7(8192) = 2656$

20,000 $\rightarrow$ virtual page no is 2 & offset is 3616

32,768 $\rightarrow$ 11 11 no is 4 offset 11 0

60,000 $\rightarrow$ 11 11 11 7 offset 2656